

Notes: Campbell and Cochrane (JPE, 1999)

By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior

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1 Big picture

- **Question.** Can a consumption-based asset-pricing model explain, all at once, the equity premium, the low and fairly stable risk-free rate, the procyclical movement of stock valuations, the long-horizon predictability of excess returns, and the countercyclical variation of stock-market volatility?
- **Core idea.** Start from standard power utility, but subtract a *slow-moving external habit*:

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} \delta^t \frac{(C_t - X_t)^{1-\gamma} - 1}{1-\gamma}.$$

When consumption gets close to habit in recessions, effective risk aversion rises sharply. That makes investors especially afraid of assets that do poorly in those states.

- **State variable.** The key object is the *surplus consumption ratio*

$$S_t = \frac{C_t - X_t}{C_t}.$$

Low S_t means consumption is near habit, marginal utility is very sensitive, and risk premia are high.

- **Main theoretical mechanism.** The model makes the *price of risk* countercyclical even though consumption growth itself is i.i.d. So the interesting asset-pricing dynamics come from preferences and the habit process, not from exogenous time variation in consumption moments.
- **Main quantitative result.** With a single slow state variable and i.i.d. lognormal consumption growth, the model reproduces:
 - a high equity premium with a low risk-free rate,
 - volatile and persistent price/dividend ratios,
 - long-horizon predictability of excess returns,
 - persistent, countercyclical volatility,
 - low unconditional correlation between consumption growth and stock returns.
- **Key novelty.** Investors fear stocks primarily because stocks perform badly in *recessions*, that is, periods in which consumption is low relative to habit, not because stocks are strongly tied to long-run average consumption growth.
- **Important robustness result.** The model works almost the same whether “the market” is treated as a claim to the consumption stream or as a claim to a volatile dividend stream that is only weakly correlated with consumption growth.
- **Historical result.** Feeding the model actual consumption data generates movements in the price/dividend ratio that track a substantial fraction of the broad secular and cyclical swings in U.S. stock prices over the last century.
- **What the paper changes in the literature.** Standard C-CAPM fails because smooth consumption growth seems too small to justify large and varying risk premia. Campbell and Cochrane keep consumption growth smooth, but amplify its pricing consequences through a state-dependent habit.

- **Economic takeaway.** The equity premium is not just “compensation for low average consumption growth.” It is compensation for doing badly in bad times when consumption is close to habit and investors are extremely risk-sensitive.

2 Calibration targets and empirical objects (key objects)

2.1 Two data sets

- **Postwar sample.** 1947–1995 stock returns from CRSP, the 3-month Treasury bill rate, and per capita nondurables-and-services consumption.
- **Long sample.** 1871–1993 stock and commercial-paper returns, with long-run consumption data from Campbell (1999), available from 1889–1992.
- **Simulation frequency.** The model is solved and simulated at a monthly frequency, then the artificial data are time-aggregated to annual data before comparing them to the annual moments in the data.

Why this matters. Time aggregation is not cosmetic. It turns out to be crucial for understanding why consumption growth and stock returns can look only weakly correlated in annual data even when the model is fundamentally consumption-based.

2.2 Two asset claims

The paper evaluates two objects.

- **Consumption claim.** A claim to the aggregate consumption stream. This is the cleanest mapping from the representative-agent model to asset-pricing theory.
- **Dividend claim.** A claim to dividends that follow

$$\Delta d_{t+1} = g + w_{t+1}, \quad w_{t+1} \sim \text{i.i.d. } N(0, \sigma_w^2), \quad \text{Corr}(w_{t+1}, v_{t+1}) = \rho. \quad (18)$$

The benchmark calibration uses an annualized dividend-growth volatility of about 11.2% and a low correlation $\rho = 0.2$ with consumption growth.

Interpretation. The paper deliberately stacks the deck against the consumption-based model by allowing dividends to be much more volatile and only weakly correlated with consumption. The striking result is that the asset-pricing implications remain very similar.

2.3 Moments used to discipline the model

The authors do *not* try to fit everything. The calibration targets a small set of moments:

- mean log consumption growth,
- standard deviation of log consumption growth,
- average log risk-free rate,

- persistence of the log price/dividend ratio,
- Sharpe ratio of excess stock returns.

Interpretation. The point is not to overfit the data. The point is to show that once the model is pinned down on a few basic moments, many other untargeted stock-market facts come out automatically.

2.4 Benchmark parameter values

The main annualized parameter choices are roughly:

- $g = 1.89\%$,
- $\sigma = 1.50\%$,
- $r^f \approx 0.94\%$,
- $\phi = 0.87$,
- $\gamma = 2$,
- $\sigma_w = 11.2\%$,
- $\rho = 0.2$.

These imply approximately:

- discount factor $\delta \approx 0.89$,
- steady-state surplus consumption ratio $\bar{S} \approx 0.057$,
- upper bound $S_{\max} \approx 0.094$.

Why $\bar{S}$ matters. Even though the power coefficient is only $\gamma = 2$, the local curvature of utility is

$$\eta_t = \frac{\gamma}{S_t}.$$

At the steady state, this is about $2/0.057 \approx 35$, and it rises further in recessions.

2.5 Untargeted facts the model is asked to explain

After calibration, the paper checks whether the model can explain:

- the level and volatility of excess stock returns,
- the mean and volatility of the price/dividend ratio,
- mean reversion in returns,
- long-horizon return predictability from the dividend yield,
- volatility-test decompositions,
- the low correlation of returns with consumption growth,
- the century-long history of stock-market valuation swings.

3 Model and equilibrium objects (all equations + interpretation)

3.1 Preferences and surplus consumption

Agents maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \delta^t \frac{(C_t - X_t)^{1-\gamma} - 1}{1-\gamma}. \quad (1)$$

Define the surplus consumption ratio

$$S_t \equiv \frac{C_t - X_t}{C_t}, \quad s_t \equiv \ln S_t. \quad (2)$$

The local curvature of utility is

$$\eta_t \equiv -\frac{C_t u_{cc}(C_t, X_t)}{u_c(C_t, X_t)} = \frac{\gamma}{S_t}. \quad (1)$$

Interpretation.

- In a boom, S_t is high: consumption is comfortably above habit, effective risk aversion is low, and risky assets are priced generously.
- In a recession, S_t is low: consumption is close to habit, effective risk aversion is high, and risky assets must offer high expected returns.

3.2 Habit dynamics

The log surplus consumption ratio follows

$$s_{t+1} = (1 - \phi)\bar{s} + \phi s_t + \lambda(s_t)(\Delta c_{t+1} - g), \quad (3)$$

while consumption growth itself is

$$\Delta c_{t+1} = g + v_{t+1}, \quad v_{t+1} \sim \text{i.i.d. } N(0, \sigma^2). \quad (4)$$

Interpretation.

- The state variable s_t is persistent because ϕ is close to one.
- The sensitivity function $\lambda(s_t)$ tells us how strongly a given consumption shock moves surplus consumption.
- In recessions, $\lambda(s_t)$ is large, so the same consumption shock has a much bigger effect on marginal utility and the discount factor.

This is the model's core departure from standard power utility: *risk sensitivity is state-dependent* even though consumption growth itself is not.

3.3 Marginal utility and the stochastic discount factor

Since habit is external, marginal utility is

$$u_c(C_t, X_t) = (C_t - X_t)^{-\gamma} = S_t^{-\gamma} C_t^{-\gamma}. \quad (2)$$

Therefore the intertemporal marginal rate of substitution is

$$M_{t+1} = \delta \left(\frac{S_{t+1} C_{t+1}}{S_t C_t} \right)^{-\gamma} = \delta G^{-\gamma} \exp \left\{ -\gamma [(\phi - 1)(s_t - \bar{s}) + (1 + \lambda(s_t))v_{t+1}] \right\}, \quad (5)$$

where $G = e^g$.

Interpretation.

- The stochastic discount factor has a *state-dependent conditional volatility*.
- When S_t is low, the coefficient on the consumption innovation, $1 + \lambda(s_t)$, is high. So one unit of consumption risk matters much more in bad times.

3.4 Sharpe-ratio bound

From the conditional Hansen–Jagannathan bound, the maximum conditional Sharpe ratio is

$$\max_{\text{all assets}} \frac{\mathbb{E}_t(R_{t+1}^e)}{\sigma_t(R_{t+1}^e)} = \left\{ \exp \left[\gamma^2 \sigma^2 (1 + \lambda(s_t))^2 \right] - 1 \right\}^{1/2} \approx \gamma \sigma (1 + \lambda(s_t)). \quad (7)$$

Interpretation.

- The price of risk rises in recessions because $\lambda(s_t)$ rises when S_t falls.
- This directly produces countercyclical expected excess returns and countercyclical Sharpe ratios.

3.5 Risk-free rate and why the sensitivity function is chosen the way it is

The log risk-free rate is

$$r_t^f = -\ln \delta + \gamma g - \gamma(1 - \phi)(s_t - \bar{s}) - \frac{\gamma^2 \sigma^2}{2} (1 + \lambda(s_t))^2. \quad (8)$$

The paper chooses $\lambda(s_t)$ to satisfy three goals:

- make the risk-free rate constant,
- keep habit predetermined at and near the steady state,
- make habit move nonnegatively with consumption and stay below consumption.

These requirements imply

$$\bar{S} = \sigma \sqrt{\frac{\gamma}{1 - \phi}}, \quad (9)$$

and

$$\lambda(s_t) = \begin{cases} \frac{1}{\bar{S}} \sqrt{1 - 2(s_t - \bar{s})} - 1, & s_t \leq s_{\max}, \\ 0, & s_t \geq s_{\max}, \end{cases} \quad (10)$$

with

$$s_{\max} = \bar{s} + \frac{1}{2}(1 - \bar{S}^2). \quad (11)$$

Substituting this choice back into the risk-free-rate formula gives the constant

$$r^f = -\ln \delta + \gamma g - \frac{\gamma}{2}(1 - \phi). \quad (12)$$

Interpretation.

- This is the paper's most "engineered" step, but it is also the cleanest one.
- The same mechanism that raises risk premia in bad times would normally also create huge movements in the risk-free rate.
- Campbell and Cochrane deliberately choose $\lambda(s_t)$ so that precautionary-savings effects offset intertemporal-substitution effects, leaving the risk-free rate roughly flat while expected excess returns move a lot.

3.6 Pricing a claim to consumption

For the consumption claim,

$$1 = \mathbb{E}_t[M_{t+1}R_{t+1}], \quad R_{t+1} = \frac{P_{t+1} + C_{t+1}}{P_t}. \quad (3)$$

Therefore the price/consumption ratio solves

$$\frac{P_t}{C_t}(s_t) = \mathbb{E}_t \left[M_{t+1} \frac{C_{t+1}}{C_t} \left(1 + \frac{P_{t+1}}{C_{t+1}}(s_{t+1}) \right) \right]. \quad (17)$$

Interpretation.

- There is a single state variable, s_t .
- Low S_t means discount rates are high, so the price/consumption ratio is low.
- Because s_t is persistent, valuations move slowly and return predictability becomes long-lived.

3.7 Pricing a claim to dividends

For the dividend claim, the recursion is

$$\frac{P_t}{D_t}(s_t) = \mathbb{E}_t \left[M_{t+1} \frac{D_{t+1}}{D_t} \left(1 + \frac{P_{t+1}}{D_{t+1}}(s_{t+1}) \right) \right]. \quad (19)$$

Its realized return can be written as

$$R_{t+1} = \frac{(P_{t+1}/D_{t+1}) + 1}{P_t/D_t} \cdot \frac{D_{t+1}}{D_t}. \quad (20)$$

Interpretation.

- The discount-rate component, embodied in the price/dividend ratio, is so dominant that the dividend claim behaves much like the consumption claim.
- That is why the model still works even when dividend growth is much noisier than consumption growth and only weakly correlated with it.

3.8 Long-run equity premium logic

At horizon k , the discount factor is

$$M_{t,t+k} = \delta^k \left(\frac{S_{t+k}C_{t+k}}{S_tC_t} \right)^{-\gamma}. \quad (24)$$

Interpretation.

- Standard power utility struggles at long horizons because long-run consumption growth is too smooth.
- This model gets around that problem because the recession state variable matters even at long horizons.
- The key tail property is that S_t itself is stationary, but $S_t^{-\gamma}$ has extremely heavy lower-tail behavior. That lets the stochastic discount factor stay very volatile at long horizons even though consumption is close to a random walk.

3.9 The paper’s deepest intuition

The model is often summarized as “habit generates high curvature,” but the more useful intuition is this:

- risk premia are mainly about *recession sensitivity*, not just contemporaneous consumption covariation;
- stocks are scary because they pay poorly when S_t is low;
- those are precisely the states in which an extra unit of consumption is most valuable.

So the paper effectively reinterprets the fundamental macro risk behind equity premia.

4 Calibration and what makes the quantitative exercise credible

4.1 Why the paper is persuasive

- **The forcing process is intentionally simple.** Consumption growth is i.i.d. lognormal. The model does *not* rely on exogenous predictability, exogenous stochastic volatility, rare shocks to expected growth, or multiple latent factors.
- **The risk-free rate is held fixed on purpose.** This prevents the model from “explaining” stock-market facts with shifting real interest rates or a disguised term premium.

- **Many moments are untargeted.** The model is pinned down with a few moments and then asked to explain a long list of additional facts.
- **It works for both consumption and dividends.** That is an important robustness check because actual stocks are claims to dividends, not to aggregate consumption.
- **Time aggregation is handled explicitly.** This matters for the consumption-return correlation puzzle.

4.2 What the model matches well

- With the benchmark calibration, the model delivers an annual excess-return Sharpe ratio of about 0.43, close to the postwar target.
- The consumption-claim excess return is about 6.6% with standard deviation about 15.2%.
- The dividend-claim excess return is about 6.5% with standard deviation about 20%.
- The price/dividend ratio is volatile enough, and its persistence is realistic.
- Long-horizon predictability is strong and very close to the postwar evidence.
- The variance decomposition of the price/dividend ratio says almost all valuation variation is expected-return variation, just as in the volatility-test literature.

4.3 What makes the paper especially influential

The paper does not merely solve the Mehra–Prescott puzzle numerically. It unifies several literatures:

- the equity-premium puzzle,
- the risk-free-rate puzzle,
- return predictability,
- excess volatility,
- countercyclical volatility,
- low consumption-return correlations.

The model says these are not six separate puzzles. They are all manifestations of one object: a slow, countercyclical discount-rate state variable.

4.4 Main limitations

- **Reverse engineering.** The sensitivity function $\lambda(s_t)$ is chosen to deliver a constant risk-free rate. That is elegant, but it is also highly tailored.
- **Very low surplus consumption.** The steady-state $\bar{S} \approx 0.057$ means habit is only a few percentage points below consumption, which is quantitatively aggressive.
- **Micro risk aversion is still high.** The paper argues that this is unavoidable in representative-agent models that match aggregate facts, but it remains a real tension.

- **External habit is more tractable than internal habit.** The paper shows some robustness, but the benchmark results are strongest in the external-habit specification.
- **No fully specified production economy.** The authors argue that a linear technology closure gives the same equilibrium and suggests robustness, but the headline model is still an endowment setup.
- **Historical fit is not perfect.** The model overpredicts the Great Depression decline in valuations and misses the rise in the early 1990s.

5 Tables and figures

5.1 Figure 1: the sensitivity function and the implied habit response

Read:

- Panel (a) plots $\lambda(s_t)$ against the surplus consumption ratio.
- Panel (b) plots how contemporaneous habit responds to consumption.

What to remember:

- $\lambda(s_t)$ is decreasing in s_t .
- In recessions (low S_t), the sensitivity of marginal utility to a consumption shock is very high.
- Near the steady state, habit is approximately predetermined.

Why it matters. This figure is the whole model in one picture. It shows how the authors create a state-dependent price of risk while keeping the risk-free rate flat.

5.2 Figures 2–6: the state variable, valuations, expected returns, volatility, and Sharpe ratios

Read:

- Figure 2: stationary distribution of S_t .
- Figure 3: price/dividend ratios as functions of S_t .
- Figure 4: expected returns and the risk-free rate as functions of S_t .
- Figure 5: conditional return volatilities as functions of S_t .
- Figure 6: conditional Sharpe ratios as functions of S_t .

Key:

- The stationary distribution of S_t is negatively skewed: deep recessions are rare but important.
- Price/dividend ratios are increasing in S_t : low in recessions, high in booms.
- Expected excess returns rise sharply when S_t falls.

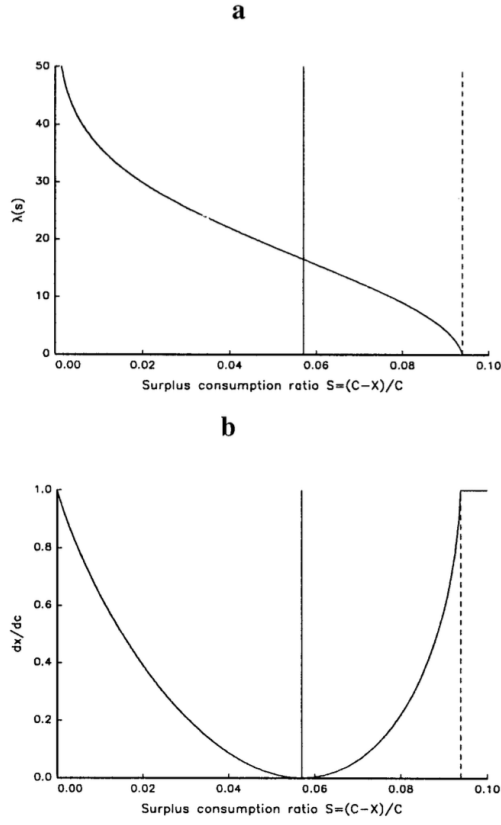


FIG. 1.—*a*, Sensitivity function $\lambda(s)$. *b*, Implied sensitivity of habit x to contemporaneous consumption. The vertical solid line in this and subsequent figures shows the steady-state surplus consumption ratio S . The dashed vertical line shows the maximum surplus consumption ratio $S_{\max}$.

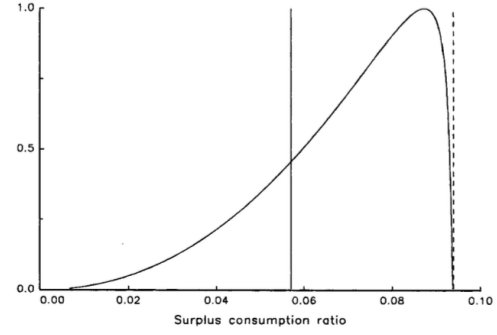


FIG. 2.—Unconditional distribution of the surplus consumption ratio. The solid vertical line indicates the steady-state surplus consumption ratio S , and the dashed vertical line indicates the upper bound of the surplus consumption ratio $S_{\max}$.

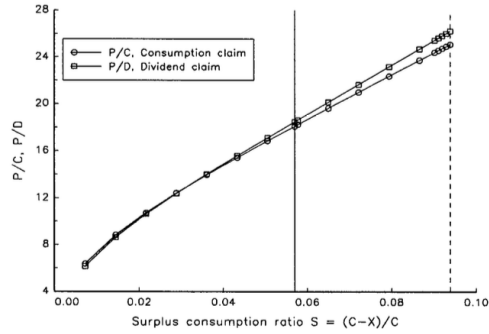


FIG. 3.—Price/dividend ratios as functions of the surplus consumption ratio

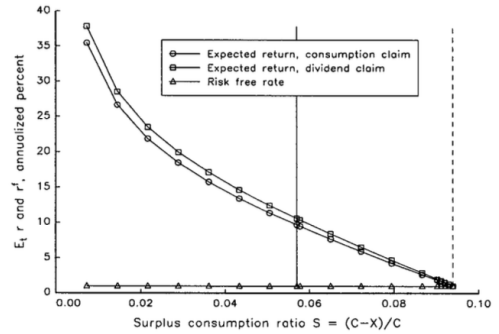


FIG. 4.—Expected returns and risk-free rate as functions of the surplus consumption ratio.

- Return volatility also rises in recessions, but less than expected returns do, so Sharpe ratios are countercyclical.
- The consumption claim and dividend claim have strikingly similar valuation and expected-return schedules.

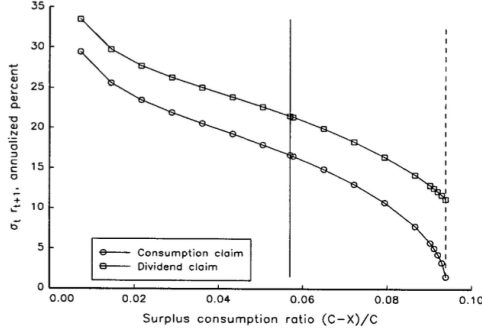


FIG. 5.—Conditional standard deviations of returns as functions of the surplus consumption ratio.

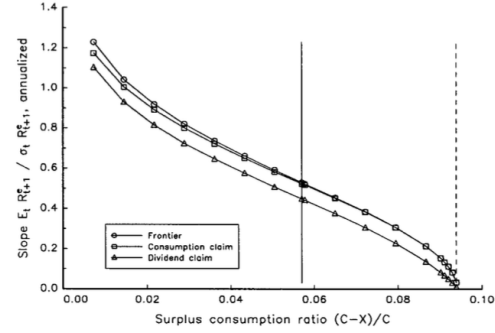


FIG. 6.—Sharpe ratios as functions of the surplus consumption ratio

Why it matters. These figures show that one slowly moving state variable generates procyclical valuation, countercyclical risk premia, countercyclical volatility, and time-varying Sharpe ratios.

5.3 Tables 1 and 2: parameter choices and first moments

Read:

- Table 1 gives the calibrated parameters.
- Table 2 compares simulated means and standard deviations with the postwar and long-run data.

TABLE 1
PARAMETER CHOICES

Parameter	Variable	Value
Assumed:		
Mean consumption growth (%)*	g	1.89
Standard deviation of consumption growth (%)*	σ	1.50
Log risk-free rate (%)*	r^f	.94
Persistence coefficient*	ϕ	.87
Utility curvature	γ	2.00
Standard deviation of dividend growth (%)*	σ_w	11.2
Correlation between Δd and Δc	ρ	.2
Implied:		
Subjective discount factor*	δ	.89
Steady-state surplus consumption ratio	$\bar{S}$	.057
Maximum surplus consumption ratio	$S_{\max}$	.094

* Annualized values, e.g., $12g$, $\sqrt{12}\sigma$, $12r^f$, ϕ^{12} , and δ^{12} , since the model is simulated at a monthly frequency.

TABLE 2
MEANS AND STANDARD DEVIATIONS OF SIMULATED AND HISTORICAL DATA

Statistic	Consumption Claim	Dividend Claim	Postwar Sample	Long Sample
$E(\Delta c)$	1.89*		1.89	1.72
$\sigma(\Delta c)$	1.22*		1.22	3.32
$E(r^f)$	.094*		.094	2.92
$E(r - r^f)/\sigma(r - r^f)$	.43*	.33	.43	.22
$E(R - R^f)/\sigma(R - R^f)$	.50		.50	
$E(r - r^f)$	6.64	6.52	6.69	3.90
$\sigma(r - r^f)$	15.2	20.0	15.7	18.0
$\exp[E(\phi - d)]$	18.3	18.7	24.7	21.1
$\sigma(\phi - d)$	.27	.29	.26	.27

NOTE.—The model is simulated at a monthly frequency; statistics are calculated from artificial time-averaged data at an annual frequency. All returns are annual percentages.

* Statistics that model parameters were chosen to replicate.

Key numbers:

- $\gamma = 2$, $\phi = 0.87$, $\delta \approx 0.89$, $\bar{S} \approx 0.057$.
- Consumption-claim excess return: about 6.6% with volatility about 15.2%.
- Dividend-claim excess return: about 6.5% with volatility about 20%.
- The model-generated Sharpe ratio is about 0.43, essentially matching the postwar target.

Why it matters. The calibrated model does not just match the Sharpe ratio mechanically; it also gets the *levels* of returns and their volatility roughly right, which is not guaranteed.

5.4 Tables 3–5: persistence, mean reversion, and long-horizon predictability

Read:

- Table 3: autocorrelations of valuation ratios, returns, and absolute returns.
- Table 4: cross-correlations between the price/dividend ratio and future returns or future volatility.
- Table 5: long-horizon return-forecasting regressions.

TABLE 3
AUTOCORRELATIONS OF SIMULATED AND HISTORICAL DATA

VARIABLE AND SOURCE	LAG (Years)				
	1	2	3	5	7
$p - d$					
Consumption claim	.87	.76	.66	.51	.39
Dividend claim	.87	.76	.66	.51	.39
Postwar sample	.87	.77	.70	.41	.04
Long sample	.78	.57	.50	.32	.29
$r - r_f^*$					
Consumption claim	-.06	-.05	-.04	-.02	-.02
Dividend claim	-.05	-.04	-.03	-.02	-.01
Postwar sample	-.11	-.28	.15	.02	.10
Long sample	.05	-.21	.08	-.14	.11
$\Sigma_{t=1}^T (p(r_t^* - r_t^*)^2)^{1/2}$					
Consumption claim	-.06	-.11	-.15	-.20	-.26
Dividend claim	-.05	-.09	-.12	-.14	-.18
Postwar sample	-.11	-.30	-.24	.18	.15
Long sample	.05	-.16	-.09	-.28	-.15
$ r_f^* $					
Consumption claim	.09	.09	.09	.07	.05
Dividend claim	.05	.05	.05	.04	.03
Postwar sample	.08	-.26	-.10	-.08	.05
Long sample	.15	.09	.07	.14	.15

Notes.—The model values are based on time-aggregated annual values with a monthly simulation interval. All data are annual.
* Partial sum of return autocorrelations out to lag j .

TABLE 4
CROSS-CORRELATIONS OF SIMULATED AND HISTORICAL DATA

VARIABLE AND SOURCE	LAG (Years)				
	1	2	3	5	7
$p - d$, r_{t+1}^*					
Consumption claim	-.35	-.30	-.26	-.20	-.15
Dividend claim	-.28	-.24	-.20	-.16	-.12
Postwar sample	-.42	-.25	-.13	-.35	-.17
Long sample	-.20	-.21	-.10	-.19	-.08
r_t^* , $ r_{t+1}^* $					
Consumption claim	-.09	-.07	-.06	-.03	-.03
Dividend claim	-.06	-.04	-.04	-.03	-.02
Postwar sample	-.32	-.14	.10	-.04	-.08
Long sample	-.15	.03	.12	.02	-.01
$p - d$, $ r_{t+1}^* $					
Consumption claim	-.49	-.42	-.37	-.28	-.21
Dividend claim	-.36	-.31	-.27	-.21	-.16
Postwar sample	-.16	.09	.11	-.05	.02
Long sample	-.12	.02	-.06	-.10	-.05

TABLE 5
LONG-HORIZON RETURN REGRESSIONS

HORIZON (Years)	CONSUMPTION CLAIM		DIVIDEND CLAIM		POSTWAR SAMPLE		LONG SAMPLE	
	10 X Coefficient	R^2	10 X Coefficient	R^2	10 X Coefficient	R^2	10 X Coefficient	R^2
1	-2.0	.13	-1.9	.08	-2.6	.18	-1.3	.04
2	-3.7	.23	-3.6	.14	-4.3	.27	-2.8	.08
3	-5.1	.32	-5.0	.19	-5.4	.37	-3.5	.09
5	-7.5	.46	-7.3	.26	-9.0	.55	-6.0	.18
7	-9.4	.55	-9.2	.30	-12.1	.65	-7.5	.23

Key numbers from Table 5 (consumption claim):

- 1-year forecasting $R^2 \approx 0.13$,
- 2-year forecasting $R^2 \approx 0.23$,
- 3-year forecasting $R^2 \approx 0.32$,
- 5-year forecasting $R^2 \approx 0.46$,
- 7-year forecasting $R^2 \approx 0.55$.

Interpretation.

- High price/dividend ratios forecast low future returns.
- Forecastability strengthens at long horizons, just as in the data.
- Returns are mean reverting because discount-rate shocks are persistent but ultimately stationary.

Why it matters. This is one of the paper's biggest victories. The standard C-CAPM has trouble with long-horizon predictability; this model makes it a natural implication of slow habit dynamics.

5.5 Table 6: variance decomposition of the price/dividend ratio

Read: Table 6 decomposes the variance of $p - d$ into the part explained by future returns and the part explained by future dividend growth.

What to remember:

- In the model, essentially all variation in $p - d$ is due to expected-return variation.
- The consumption claim assigns about 100% of $\text{Var}(p - d)$ to returns and about 1% to dividends.
- The dividend claim gives almost the same answer.

Interpretation. This is the model’s “excess-volatility” result. Prices move a lot because discount rates move a lot, not because expected cash flows move a lot.

Why it matters. The paper offers a structural explanation for the Campbell–Shiller/Cochrane result that valuation ratios mainly forecast returns rather than dividend growth.

TABLE 6
VARIANCE DECOMPOSITIONS

Source	Returns (%)	Dividends (%)
Consumption claim	100	1
Dividend claim	99	3
Postwar sample	137	−31
Long sample	101	−10

NOTE.—Table entries are the percentage of $\text{var}(p - d)$ accounted for by dividend growth and returns.

$$100 \times \sum_{j=1}^N \frac{\rho^j \text{cov}(p_t - d_t, x_{t+j})}{\text{var}(p_t - d_t)},$$

$x = -r$ and Δd , respectively.

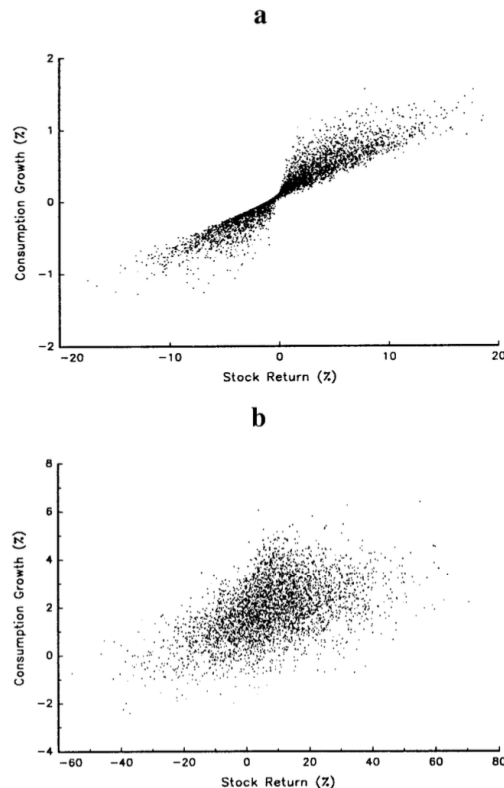


FIG. 7.—*a*, Simulated monthly consumption growth vs. monthly consumption claim returns. *b*, Growth in simulated annual consumption vs. annual consumption claim returns.

5.6 Figure 7 and Table 7: why returns and consumption growth can look weakly related

Read:

- Figure 7 plots simulated return–consumption pairs.
- Table 7 reports lead-lag correlations of returns with consumption growth.

What to remember:

- In the underlying monthly model, returns and consumption growth are highly related, but not perfectly one-for-one because the slope changes with the state variable.
- Time aggregation dramatically weakens the contemporaneous annual correlation.
- In annual data, returns become negatively correlated with past consumption growth and positively correlated with future consumption growth, matching the sign pattern in the data.

TABLE 7
CORRELATION BETWEEN CONSUMPTION GROWTH AND STOCK RETURNS

CORRELATION	MONTHLY MODEL		ANNUAL MODEL		DATA SAMPLE		
	Consumption Claim	Dividend Claim	Consumption Claim	Dividend Claim	Quarterly Postwar	Annual Postwar	Annual Long
$r_{\Delta c, \Delta c}^s$	.00	.00	-.16	-.15	-.05	-.16	-.05
$r_{\Delta c, \Delta d}^s$	.00	.00	-.19	-.15	-.10	-.34	-.08
$r_{\Delta c, \Delta c}^a$	.93	.79	.47	.40	.12	-.05	.09
$r_{\Delta c, \Delta d}^a$	.00	.00	.50	.42	.19	.37	.49
$r_{\Delta d, \Delta d}^a$	.00	.00	.00	.00	.15	-.26	.05

Important number. The contemporaneous annual correlation between consumption-claim returns and annual consumption growth drops to about 0.47 in the model, much lower than the near-perfect relation implied by standard time-separable models.

Why it matters. This helps explain why empirical C-CAPM tests often find that consumption growth looks like a weak pricing factor even if the true stochastic discount factor is fundamentally consumption-based.

5.7 Table 8: returns are better proxies for the discount factor than consumption growth

Read: Table 8 reports correlations of the true stochastic discount factor with consumption growth, the consumption-claim return, and the dividend-claim return.

Key numbers:

- Monthly correlations: about 0.90 with consumption growth, 0.99 with the consumption-claim return, and 0.83 with the dividend-claim return.
- Annual correlations: about 0.45 with consumption growth, 0.99 with the consumption-claim return, and 0.80 with the dividend-claim return.

Interpretation. Even in a model where consumption is the true pricing kernel, stock returns can be better empirical proxies for the discount factor than measured consumption growth.

Why it matters. This helps reconcile the success of return-based factor models with the deeper logic of consumption-based pricing.

TABLE 8
CORRELATION OF THE STOCHASTIC DISCOUNT FACTOR WITH
CONSUMPTION GROWTH, CONSUMPTION CLAIM RETURN,
AND DIVIDEND CLAIM RETURN

	CORRELATION OF STOCHASTIC DISCOUNT FACTOR WITH:		
	Consumption Growth	Consumption Claim Return	Dividend Claim Return
Monthly	.90	.99	.83
Annual	.45	.99	.80

NOTE.—The stochastic discount factor is

$$M_{t+1} = \delta \left(\frac{C_{t+1} S_{t+1}}{C_t S_t} \right)^{\gamma}$$

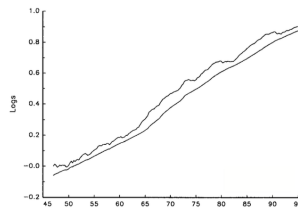


FIG. 8.—Nondurable and services consumption per capita and habit level implied by the model, under the assumption that the surplus consumption ratio starts at the steady state.

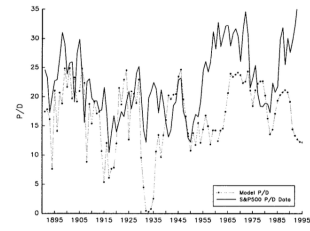


FIG. 9.—Historical price/dividend ratio and model predictions based on the history of consumption.

5.8 Figures 8 and 9: feeding the model actual consumption data

Read:

- Figure 8 shows actual consumption together with the habit series implied by the model.
- Figure 9 compares the model-implied price/dividend ratio, based only on historical consumption data, with the actual stock-market price/dividend ratio.

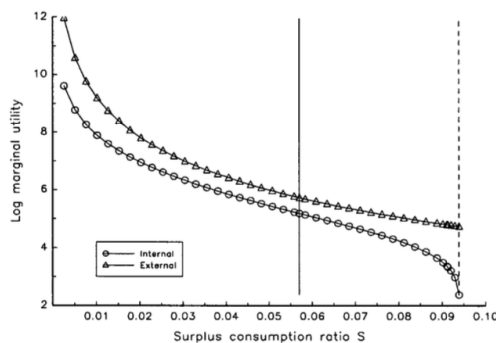


FIG. 10.—Marginal utility with internal vs. external habits. In each case, the marginal utility of consumption is given by $MU_t = C_t^{-1} f(S_t)$; the figure plots $f(S_t)$.

What to remember:

- Habit moves smoothly and slowly.
- When consumption dips toward habit, the model predicts low valuations and high expected returns.
- The model captures many major secular and cyclical swings: late nineteenth-century weakness, the 1920s boom, the Depression collapse, postwar valuation movements, and the 1970s decline.

Where it misses.

- It predicts too severe a valuation collapse in the Great Depression.
- It misses the run-up in the early 1990s.

Why it matters. The authors are not merely matching moments. They show that the model, fed only with realized consumption, can tell a historically recognizable story about the stock market.

5.9 Figure 10 and Section IV: micro implications and robustness

Read: Figure 10 compares marginal utility under external and internal habit.

What to remember:

- Internal- and external-habit marginal utilities are nearly proportional near the steady state.
- Many asset-pricing implications are therefore robust near the center of the state space.
- But external habit produces stronger variation in expected returns and predictability in the tails.

Other micro results from Section IV:

- The model can accommodate heterogeneity through reference groups and appropriately scaled individual endowments.
- Micro risk aversion is still high: the paper reports risk aversion around 80 at the steady state and higher in recessions.

Why it matters. The paper is honest about the remaining tension between matching aggregate asset prices and fitting micro evidence.

6 Short summary

- **Method.** Take standard consumption-based pricing, add a slow-moving external habit, and let the surplus consumption ratio be the single persistent state variable.
- **Main conceptual result.** Time variation in expected returns can come from state-dependent risk aversion rather than from state-dependent consumption growth moments.
- **Main quantitative result.** The model matches the equity premium, a low risk-free rate, volatile and persistent valuation ratios, long-horizon return predictability, and countercyclical volatility.
- **Main empirical interpretation.** Valuation ratios move mainly because discount rates move, not because expected dividend growth moves.
- **Main macro-finance intuition.** Investors primarily fear recession states in which consumption is close to habit.

7 Conclusion

Core conclusion. Campbell and Cochrane build one of the most successful representative-agent asset-pricing models in macro-finance by making the marginal utility of consumption depend on *consumption relative to habit*, not just consumption itself.

Economic intuition. In good times, investors are comfortable and require modest compensation for bearing risk. In bad times, when consumption approaches habit, the same payoff risk becomes extremely painful. This makes expected excess returns rise sharply in recessions while the risk-free rate can remain roughly unchanged.

Why the paper lasts. The paper gives a unified explanation for several classic asset-pricing facts that used to be treated separately. It also changes the language of macro-finance: after this paper, it becomes natural to think about stock returns as compensation for doing badly in *surplus-consumption recessions*.

Takeaway. The model's central contribution is not merely that habit can raise the equity premium. It is that a slow-moving recession state, embedded in preferences, can generate realistic discount-rate dynamics and thereby explain a broad cross-section of time-series facts about aggregate stock prices.